

Q1. Area in terms of range

Let the flight time be t . Resolving the initial velocity,

$$\vec{v}_0 = (v_0 \cos \theta, v_0 \sin \theta), \quad \vec{g}t = (0, -gt).$$

The cross-product magnitude of any two sides of the triangle gives (twice) its area. Choosing $\vec{v}_0$ and $\vec{g}t$,

$$2S = |\vec{v}_0 \times (\vec{g}t)| = |(v_0 \cos \theta)(-gt) - (v_0 \sin \theta)(0)| = gtv_0 \cos \theta.$$

Since $x = v_0 \cos \theta \cdot t$ is precisely the horizontal range, hence

$$S = \frac{1}{2}gx.$$

Q2. Area in angular form

For ground-to-ground motion, the vertical displacement is zero:

$$0 = v_0 \sin \theta \cdot t - \frac{1}{2}gt^2 \implies t = \frac{2v_0 \sin \theta}{g}.$$

Substituting into the area formula of Q1,

$$S = \frac{1}{2}g(v_0 \cos \theta) \frac{2v_0 \sin \theta}{g} = \boxed{\frac{1}{2}v_0^2 \sin 2\theta}.$$

Now consider the same triangle but with the two sides $\vec{v}_0$ and $\vec{v}$ meeting at the origin. For ground-to-ground motion, energy conservation gives $|\vec{v}| = |\vec{v}_0| = v_0$. The (signed) area of the triangle is

$$S = \frac{1}{2} |\vec{v}_0| |\vec{v}| \sin \alpha = \frac{1}{2} v_0^2 \sin \alpha,$$

where α is the interior angle at the origin, taken from $\vec{v}_0$ to $\vec{v}$. At landing $\vec{v} = (v_0 \cos \theta, -v_0 \sin \theta)$, so $\alpha = -2\theta$ (the velocity has rotated clockwise by 2θ). Hence

$$\sin \alpha = \sin(-2\theta) = -\sin 2\theta, \quad S = -\frac{1}{2}v_0^2 \sin \alpha.$$

The two forms are therefore equivalent: $S = \frac{1}{2}v_0^2 \sin 2\theta = -\frac{1}{2}v_0^2 \sin \alpha$.

The maximum of S — and therefore of x — occurs when $\sin 2\theta = 1$, i.e. $\theta = 45^\circ$, recovering the familiar result through a purely geometric route.

Q3. Non-zero launch height

The mass is launched from $y = h > 0$ and lands at $y = 0$. The flight time satisfies

$$0 = h + v_0 \sin \theta \cdot t - \frac{1}{2}gt^2,$$

giving

$$t(\theta) = \frac{v_0 \sin \theta + \sqrt{v_0^2 \sin^2 \theta + 2gh}}{g}.$$

The horizontal range is $L(\theta) = v_0 \cos \theta \cdot t(\theta)$.

To maximise L , set $dL/d\theta = 0$. Writing L explicitly and differentiating (or, more elegantly, using the implicit relation $L \tan \theta - gL^2/(2v_0^2 \cos^2 \theta) + h = 0$ obtained by eliminating t), one finds

$$\boxed{\sin^2 \theta_{\text{opt}} = \frac{v_0^2}{2(v_0^2 + gh)}}.$$

Equivalently,

$$\theta_{\text{opt}} = \arcsin \sqrt{\frac{v_0^2}{2(v_0^2 + gh)}}, \quad \text{or} \quad \tan \theta_{\text{opt}} = \frac{v_0}{\sqrt{v_0^2 + 2gh}}.$$

Geometric note. From Q1, $S = \frac{1}{2}gL$ still holds. Energy conservation gives $|\vec{v}|^2 = v_0^2 + 2gh$, so the two sides meeting at the origin have *fixed* lengths v_0 and $\sqrt{v_0^2 + 2gh}$. The area $S = \frac{1}{2}v_0 |\vec{v}| \sin \varphi$ (where φ is the angle between $\vec{v}_0$ and $\vec{v}$) is maximised when $\varphi = 90^\circ$, i.e. when $\vec{v}_0 \perp \vec{v}$. Imposing orthogonality on the vector triangle yields the same optimal angle as above, without calculus.